

poignant ending that the two stories share: a well-meaning father who, having no inkling of the distress he's about to cause, disillusion his hopeful child by revealing that the child's visual analogy, though seemingly a gold nugget, is in fact just a piece of fool's gold. All in all, there are a large number of elements common to both stories, the most crucial one being a hopeful child's sudden experience of keen disappointment.

Much like the case of Danny playing with ants on the ground and Dick picking up bottlecaps from the ground, this case shows that abstract reminders aren't triggered solely by a close matchup between the most abstract cores of the two events, but that matchups at several levels of abstraction are often needed to bring them about. Obviously many details are irrelevant, such as the colors of the buttons on the Dustbuster, the ages of the children, the houses where the events took place, and so on, but one aspect that is not irrelevant is the fact that in both stories, it was a parent — specifically, the father — who disillusioned a child. Although this is a superficial aspect of both stories (as is the dirt in the Danny and Dick stories), it certainly contributes to their striking parallelism. And there's no doubt that the analogy is rendered yet more intriguing by the fact that little Doug, the “disillusionee” of the first story, grew up to be the *disillusioner* in the second story. To anyone who hears both stories and sees their many analogical connections, this role-reversal has to be one of the most central and ironic aspects of the whole thing, and it no doubt strengthens the analogy, even though the younger and older Doughts play opposite roles in their respective stories.

In sum, several emotions — an initial fascination, the pleasant surprise of a simple visual analogy, the high hopes that ensue from this discovery, and finally a sudden deflation — all play critical roles in this reminding. This illustrates a general tendency — namely, that reminders that take place at a deep level are often dependent on emotional aspects of the two episodes they link together. This is because very often the most central aspects of an event are the strong emotions that it churns up.

Events are Encoded Not by Rote but by Distillation

So far, we've been using the term “encoding” as if what we meant by it were self-evident. Our physiology restricts our perception to the standard sensorial modalities (vision, hearing, etc.) and to the features afforded by those modalities (colors, movements, and shapes, for example); our perception is also constrained by the resolving power of our senses (the world would appear very different to us if our visual system could directly perceive microbes). Our psychology also limits our perception, allowing us to recognize and memorize events only in terms of certain modalities of encoding. We perceive through our sensory organs, to be sure, but no less through our concepts; in other words, we perceive not just physiologically but also intellectually. There is thus an unbreakable link between perceiving and conceiving. On the one hand, our conceptions depend on our senses, since our concepts would be quite different if our senses were different, but on the other hand, our perceptions depend on our repertoire of concepts, because the latter are the filters through which any stimulus in our environment reaches our consciousness.

It may not be obvious why we need to encode our experiences at all — that is, why we need to reduce them to a tiny fraction of their entirety. To see why encoding is necessary, imagine trying to memorize an event without any simplification taking place; the result might be called a “total rote recording” or “perception without concepts”. An experience would be captured in its entirety in our neurons, much as a film can be stored on a DVD. In the case of Danny and the Grand Canyon, having such a “total rote recording” would mean that the entire scene had been “filmed” in Doug’s brain while he was experiencing it, and then that, some twenty years later, when he observed Dick stooping to pick up a bottle cap, this specific film had been reactivated in his brain by a mental search algorithm running through all filmed scenes in his memory.

This idea of “perception without concepts” can be summarily rejected, because no mental process based solely on visual cues would be able to connect one-year-old Danny with the mature adult Dick, or to see the link between ants and a bottlecap, not to mention the link between the enormous geological concavity of the Grand Canyon and the architectural and archeological masterwork of the Temple at Karnak. A search for purely *visual* resemblances, based on techniques having to do with alignment of images, could not possibly lead to such a reminding. And so far we have only alluded to the *physical entities* in the scene, while completely leaving out the *actions* (manipulating small items on the ground), the *context* (a two-week-long pleasure trip), the *relationships* (the contrasting sizes of the entities involved in the situations), the *emotions* (a feeling of irony), and so forth.

If conceptual encoding did not take place, it would be impossible to retrieve events stored in one’s memory. Just think how indispensable it is for users of Web sites on which photos or videos are shared that linguistic labels (“tags”) are attached to each item. There are many additional arguments for the necessity of encoding in the act of committing experiences to memory, based on such phenomena as selective forgetting, partial or distorted recovery of memories, and the reconstruction of memories.

No purely image-based search process, no matter how sophisticated, would be able to “see” the connection between Dick at Karnak and Danny at the Grand Canyon, or between Doug’s disappointment about subscripts and Monica’s disappointment about button #2, because such events’ connections are not visual. The moral is that we do not store in our memory a collection of “objective” events through which we run, seeking perceptual resemblances, whenever a new event happens to us; rather, events that befall us get encoded — that is, perceived, distilled, and stored — in terms of prior concepts that we have acquired. Reminders are possible because certain aspects of long-ago situations were noticed and stored at that time, yielding encodings of those situations. But which aspects get paid attention to, and how are they encoded?

Do Our Brains Instantly Pinpoint Timeless Essences?

The activation of certain memories from one’s earlier life is not a mental game that we humans engage in merely because we find it intellectually pleasurable to connect present and past through the spotting of similarities. In general, the automatic

behaviors that we engage in play a crucial role in ensuring our survival. Being reminded of a past event is not a luxury add-on that might optionally take place after we have understood a new event; rather, such a reminding is deeply implicated in the very act of understanding the new situation. This idea was stressed early on by cognitive scientist Roger Schank, one of the pioneers in the study of reminding.

Let us restate this more concretely. Doug was reminded of his son Danny at the Grand Canyon when he watched his friend Dick at Karnak because his perception of the fresh new Karnak situation activated certain concepts that had been encoded in the course of “processing” the Grand Canyon event many years earlier. Doug knew he was in a very special sacred place when he was at Karnak. He noticed that a companion had leaned down to pay attention to something small and was momentarily ignoring the guide. His emerging sense of irony in this situation led him to “replay” the scenario of Danny at the Grand Canyon, as it existed in his memory.

Does this mean that the category *trivial side show more fascinating than the main event* was necessarily created at the moment the first event was encoded? Does it mean that Doug, when he experienced the second event and encoded it, automatically reactivated the category’s founding member because the two had been encoded *identically*? This is the idea on which Schank’s theory of the mechanisms underlying reminding is based. However, is that the only way to explain the bubbling-up of a dormant memory?

Does a successful reminding presume that both events were encoded at such a high level of abstraction that the two situations are simply specific cases of the abstraction? Can one be reminded of a past situation only if, already in the moment when one was experiencing it, one succeeded in putting one’s finger on such an abstract “timeless essence” that later, many years down the pike, when another situation sharing that same timeless essence comes along, one will be all prepared to activate it? If this were the case — let’s call it “instant pinpointing of timeless essences” — then Doug, when he observed Danny’s fascination with the ants and the leaves, would have instantly created, albeit unconsciously and without an explicit linguistic label, the abstract category *trivial side show more fascinating than the main event*. His sudden recollection, as he watched Dick at Karnak, of his son at the Grand Canyon would have been due to the creation of this category some fifteen years earlier. If one were to believe the hypothesis of “instant pinpointing of timeless essences”, then such a reminding could have taken place only if the conceptual skeleton *trivial side show more fascinating than the main event* (we stress that it’s not the sequence of English words that we are talking about, but the abstract idea that it denotes) had been created at the moment of encoding the first scene, and then, fifteen years later, rediscovered in another scene.

To put it mildly, this is most implausible. The instant spotting of a shared timeless essence is not the only mechanism that could, in theory, give rise to a reminding sometime down the pike. The problem is not that it’s hard to perceive a deep and precise conceptual skeleton, for in fact we all perceive such skeletons all the time, when the categories involved are those of our day-in-day-out thinking — that is, categories with which we have some degree of expertise. The problem is that when the categories are unfamiliar to us, we cannot see nearly as deeply into situations involving them.

We effortlessly and unconsciously perceive certain shared aspects of situations that are very different — for example, despite their enormous differences, we recognize all sorts of *elephant-in-the-room* situations, *once-bitten-twice-shy* situations, *you’re-pushing-your-luck* situations, *can’t-see-the-forest-for-the-trees* situations, *if-it-ain’t-broke-don’t-fix-it* situations, *killing-two-birds-with-one-stone* situations, *a-bird-in-the-hand-is-worth-two-in-the-bush* situations, *you-can’t-have-your-cake-and-eat-it-too* situations, *you-could-hear-a-pin-drop* situations, *people-who-live-in-glass-houses-shouldn’t-throw-stones* situations, and so forth — and the recognition of these highly abstract qualities allows us to spot commonalities in situations that have nothing at all in common on their surfaces. In a word, we are all experts at perceiving conceptual skeletons when the categories involved are ones in which... we are experts!

And so, could it be that Doug unconsciously perceived the conceptual skeleton *trivial side show more fascinating than the main event* while watching Danny, and used it to encode that event? Could it be that this made him an expert with this concept? Could it be that thanks to his expertise with this concept he was able to recall that event many years later, while visiting Karnak? Well, it’s possible in theory, but it’s most unlikely.

Spotting the essence of something that one is already an expert in is one thing, but spotting the essence of something novel and unfamiliar with is quite another thing. To put it more pithily, abstraction is one thing, but *deep* abstraction is quite another thing. The hypothesis that we can instantly spot “timeless essences” leads, in certain cases, to absurd conclusions. When experiencing an event, we cannot peer into the future and clairvoyantly guess exactly which highly abstract encoding to construct so as to allow that memory to be triggered by events that will take place many years down the pike.

More concretely put, when Doug was eight years old and he asked his father what the meaning of subscripts was and his father’s answer gave rise to a great feeling of loss, the abstract conceptual skeleton that this event would wind up sharing, forty years later, with the episode of Monica and the Dustbuster could not possibly have been clear to him at that early point in his life. What, then, *did* the eight-year-old child encode at that time? Alas, one would need both a time machine and a mind-reading machine to give a precise answer, but there is no doubt that *some* abstraction was involved in the encoding. The crux of perception — even for an eight-year-old or a two-year-old — is the act of abstracting; abstraction is the principle that allows us to create new categories and to extend them throughout the course of our lives. We are forever extending our categories because the strictest form of literality (“total rote recording”, as we dubbed it above) does not allow *any* resemblances to be noticed, and thus excludes all thinking. In the case of eight-year-old Doug, there had to be at least enough abstraction in his perception to allow the connection with the Dustbuster scenario forty years later to be perceived, even though that scenario was, in so many ways, vastly different from it.

First Encodings Can Go Only So Far

In a course one of us taught, a student heard the anecdote of newlywed Carol signing with her maiden name and her friend Peter who “did the same thing every January”; this reminded the student of a time four years earlier, when she had changed